Stochastic Finance and Risk Modelling

14/22 ≈ 19/20

FINAL EXAM

2H, Neither Document nor Calculator is Allowed

The final score takes values between 0 and 20, while the accumulated points are up to 27. The scale is indicative only.

Exercise 1. "Chooser" Option 4 pt

We consider a finite discrete-time financial market consisting of a *risk-free asset*, whose price process S^0 is given by $S_n^0 = e^{rn}$ for all $n \in \mathbb{N}$ with $r \geq 0$, and one risky asset with price process $S = (S_n)_{n\in\mathbb{N}}$. The filtration is $\mathbb{F} = (\mathcal{F}_n)_{n\in\mathbb{N}}$, with $\mathcal{F}_n := \sigma(S_0, \ldots, S_n)$. We assume that the market is arbitrage-free and complete.

We are interested in a "chooser" option with maturity N and strike price K. The particularity of this option is that the buyer will choose at a predetermined date $T_c \in \{1, \ldots, N\}$ whether the option is a call or a put ; in both cases, the maturity of the options is N, and the strike is K. Let $C_n \equiv C_n(K, N)$ (resp. $P_n \equiv P_n(K, N)$) be the arbitrage-free price at time n of the call (resp. the put). The payoff at N is given by :

4

$$(S_N - K)_+ \mathbb{1}_{C_{T_c} \geq P_{T_c}} + (K - S_N)_+ \mathbb{1}_{C_{T_c} < P_{T_c}}.$$

1. Recall the Fundamental Theorem of Asset Pricing in this complete arbitrage-free market.
2. Show that the arbitrage-free price of the "chooser" option at T_c is $\max(C_{T_c}, P_{T_c})$.
3. Using the put-call parity relation, show that the arbitrage-free price at T_c of this option is :
$$C_{T_c} + \left(e^{-r(N-T_c)}K - S_{T_c}\right)_+.$$
4. Deduce that the "chooser" option can be replicated before T_c by a portfolio consisting of a call and a put, specifying their respective maturities and strike prices.
5. In terms of $C_0(\cdot,\cdot)$ and $P_0(\cdot,\cdot)$, what is the arbitrage-free price of the "chooser" option at 0 ?

Exercise 2. American/European Option 4 pt

This exercise aims to prove that an American call option has the same price as a European call option. We consider a finite discrete time probability model $(\Omega, \mathcal{F}, \mathbb{P})$ with a time horizon N and a filtration $\mathbb{F} = \{\mathcal{F}_n,\ n \geq 0\}$ where $\mathcal{F}_0 = \{\emptyset, \Omega\}$ and $\mathcal{F}_N = \mathcal{F}$.
There are $d+1$ assets, the first indexed by 0, called the *risk-free asset*, whose price process S^0 is given, for all $n \in \{0, \ldots, N\}$, by $S_n^0 = (1+r)^n$, with $r \geq 0$.
The next d assets are called *risky assets*, and their price processes, denoted S^i, $i = 1, ..., d$, are

assumed to be $\mathbb{F}$-adapted.
We assume the market is arbitrage-free and complete. Let $Z = (Z_n; n = 0 \dots N)$ be an adapted process. We denote by $\mathcal{C}_n$ the arbitrage-free value at time n of the American option with payoff process Z. Recall that the holder can exercise the American option at any time $n \in \{0, \dots, N\}$. If the option is exercised at n, the holder will receive a payoff equal to Z_n. The option can only be exercised once. Let c_n be the arbitrage-free value at time n of the European option with $\mathcal{F}_N$-measurable payoff defined by $h = Z_N$.

1. Recall the recursive formulation of $\mathcal{C}_n$.
2. Show that $\mathcal{C}_n \geq c_n$, for all $n = 0 \dots N$.
3. Show that if $c_n \geq Z_n$, for all $n = 0 \dots N$, then $\mathcal{C}_n = c_n$, for all $n = 0 \dots N$.
4. Consider a market with a single risky asset with price process S and apply the previous result to the case of a European call option on S with maturity N and strike price K.

Exercise 3. Optimal Portfolio 6 pt

Consider the nonlinear optimization problem :

$$\begin{aligned} w_\phi^* &:= \operatorname{argmax}_{w \in \mathbb{R}^n} \left(\mu^T w - \frac{\phi}{2} w^T \Sigma w \right) \\ &\text{s.t. } \mathbf{1}^T w = 1, \end{aligned} \tag{1}$$

where $\mu \in \mathbb{R}^n$ is the vector of returns of the n risky assets, Σ is the $n \times n$ covariance matrix, $\mathbf{1} := (1, \dots, 1) \in \mathbb{R}^n$, and $\phi > 0$ is the risk aversion parameter. We assume that Σ is invertible and positive semi-definite and that μ is not collinear to $\mathbf{1}$. Let $a = \mu^T \Sigma^{-1} \mathbf{1}$, $b = \mu^T \Sigma^{-1} \mu$ and $c = \mathbf{1}^T \Sigma^{-1} \mathbf{1}$.

1. Compute the optimizer w_ϕ^* as a function of a, b, c and Σ, μ, and ϕ.
2. Deduce the limit $\lim_{\phi \to \infty} w_\phi^*$.
3. Compute the return and the variance of w_ϕ^*.
4. Prove that $bc - a^2 > 0$ and compare the return of w_ϕ^* to a/c.
5. Express the variance of w_ϕ^* as a function of its return. What do you deduce ?

Exercise 4. Markov Process and Average Options 7 pt

The binomial market consists of two assets : The first one is called the *risk-free asset*, and its price process, S^0, is given, for all $n \in \{0, \dots N\}$, by $S_n^0 = (1 + r)^n$, with $r > -1$.
The second one is called the *risky asset*, and its price process, S, is given by :

$$\begin{aligned} S_{n+1} &= S_n U_{n+1}, \quad \text{for } n \in \{0, \dots N-1\} \\ S_0 &= s \in \mathbb{R}_*^+. \end{aligned}$$

The random variables U_n are all defined on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$, taking values in $\{d, u\}$ with $0 < d < u$, that is,

$$\begin{aligned} U_n : \Omega = \{u, d\}^N &\longrightarrow \{u, d\} \\ \omega = (\omega_n)_{n \in \{1, \dots N\}} &\mapsto \omega_n \end{aligned}$$

We set $\mathbb{F} = \{\mathcal{F}_n,\ n \geq 0\}$ with $\mathcal{F}_0 = \{\emptyset, \Omega\}$ and $\mathcal{F}_n = \sigma(U_1, \dots, U_n)$, for $n \in \{1, \dots N\}$.
We assume that $(\Omega, \mathcal{F}, \mathbb{P})$ is non-redundant, meaning that $\mathcal{F} = \mathcal{P}(\Omega)$ and for all $\omega \in \Omega$,

$\mathbb{P}(\omega) > 0$.

In a generic way, we denote by $\tilde{A}$ the normalized process of A, i.e., $\tilde{A}_n = \dfrac{A_n}{S_n^0}$ for all $n \in \{0, \dots N\}$.

I) Introduction

1. Show that $\mathcal{F}_n = \sigma(S_1, \dots, S_n)$. What does $\mathcal{F}_n$ represent ?
2. Provide a necessary and sufficient condition for the market to be arbitrage-free and complete. *We will assume that this condition is satisfied from now on.*
3. Define $\mathbb{P}^*$, so that it is the unique probability measure that makes $\tilde{S}$ a martingale. Construct $\mathbb{P}^*$ from a certain real number π that you will express in terms of u, d, and r. What can we say about $(U_n; n \in \{1, \dots, N\})$ under $\mathbb{P}^*$? *Questions I2) and I3) are questions about the course ; no proof is required.*
4. Let g be a Borel measurable and integrable function from $\mathbb{R}^{n+1}$ to $\mathbb{R}$. Prove using the definition of the conditional expectation that

$$\begin{aligned} \mathbb{E}^* \left(g(S_1, \dots, S_n, U_{n+1}) | \mathcal{F}_n\right) &= G(S_1, \dots, S_n) \text{ , where} \\ G(s_1, \dots, s_n) &= \mathbb{E}^* \left(g(s_1, \dots, s_n, U_{n+1})\right). \end{aligned}$$

II) Option Pricing We introduce the process $\Sigma = (\Sigma_n; n \in \{0, \dots N\})$, defined as :

$$\Sigma_n = \sum_{j=1}^{n} S_j.$$

1. Let h be a Borel measurable and integrable function from $\mathbb{R}^2$ to $\mathbb{R}$. Show that, for all $n = 0, \dots N-1$,

$$\mathbb{E}^* \left(h(S_{n+1}, \Sigma_{n+1}) | \mathcal{F}_n\right) = \pi h(uS_n, \Sigma_n + uS_n) + (1-\pi) h(dS_n, \Sigma_n + dS_n).$$

2. Deduce from 1) that $C = (C_n = (S_n, \Sigma_n); n \in \{0, \dots N\})$ is a $(\mathbb{F}, \mathbb{P}^*)$-Markovian process.
3. We now consider average options whose terminal payoff is of the form $e(S_N, \Sigma_N)$. For example, if $e(x, y) = (x - \dfrac{y}{N+1})^+$, this corresponds to an average strike call option maturing at N. We denote by E_n the arbitrage-free value at date n of the option with payoff $e(S_N, \Sigma_N)$ at date N. Explain what the arbitrage-free value E_n is and show that there exists, for all $n = 0, \dots N$, a function e_n from $\mathbb{R}^2$ to $\mathbb{R}$ such that $E_n = e_n(S_n, \Sigma_n)$, where

$$\begin{aligned} e_N &= e \\ e_n(x, y) &= \frac{1}{1+r} \left[\pi e_{n+1}(ux, ux + y) + (1-\pi) e_{n+1}(dx, dx + y)\right] \quad n = 0, \dots, N-1. \end{aligned}$$

Exercise 5. Risk Management 6 pt

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. We consider a market with a time horizon of 1 and with two assets. The risky asset S is such that $S_0 = s$ for a given constant $s \in \mathbb{R}$, we call $\Delta S = S_1 - s$ and assume that ΔS is square integrable under $\mathbb{P}$. We also assume that the riskless asset is such that $S^0 \equiv 1$. We consider a derivative with payoff $F(S_1)$, and we want to find the optimal hedging of $F(S_1)$ from the point of view of risk management. We call p the price of $F(S_1)$ observed in the market. We assume that $F(S_1)$ is also square integrable under $\mathbb{P}$. Define the risk measure ρ for the square-integrable random variables X by

$$\rho(X) := \mathbb{E}[X^2].$$

2

1. Let $x, y \in \mathbb{R}$. Consider a strategy where there is x in S, y in S^0 and -1 in $F(S_1)$. Let V be the associated portfolio value. Write V_0, V_1 and further the loss $L := V_0 - V_1$. We say that (x, y) is a hedging strategy if $p = xs_0 + y$. Compute the loss in this case. We write it $L(x, p)$.
   We want to find among (x, p) the one that minimize
$$\min_{x,p} \rho(L(x,p)).$$
2. Write the optimality conditions for x, p. Solve the system and show that the optimal solution p^* is given by
$$p^* = \frac{\mathbb{E}[\Delta S^2]\mathbb{E}[F(S_1)] - \mathbb{E}[\Delta S]\mathbb{E}[F(S_1)\Delta S]}{\mathrm{var}(\Delta S)},$$
   where $\mathrm{var}(\Delta S)$ is the variance of ΔS.
3. Assume that S is a martingale under $\mathbb{P}$. Compare p^* to the arbitrage-free price of $F(S_1)$.
4. Assume in what follows that we are under the setting of the binomial model where $(0,1) \ni \mathbb{P}[S_1 = (1+u)s] = 1 - \mathbb{P}[S_1 = (1+d)ds]$, where $u > 0 > d > -1$. Compute p^* in this context and show that
$$p^* = -\frac{d}{u-d}F((1+u)s) + \frac{u}{u-d}F((1+d)s).$$
   Hint : if $q = \mathbb{P}[S_1 = (1+u)s]$, $\mathrm{var}(\Delta S) = s^2 q(1-q)(u-d)^2$.
   Compare again p^* to the arbitrage-free price of $F(S_1)$
5. Now prove that there exists $(\bar{x}, \bar{p})$ such that $\rho(L(\bar{x}, \bar{p})) = 0$. Comment.